Integrative Nutrient Pattern Clustering and Multivariate Metabolic Risk Stratification in U.S. Adults: Evidence from NHANES 2017–2018

Advanced Research Portfolio — Automated Manuscript

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1. Abstract

Background: Integrating national dietary surveys, reference food composition repositories, and bibliometric corpora enables hypothesis-rich modeling of diet–health relationships with explicit provenance. Methods: We harmonized NHANES dietary total nutrient files with demographic and examination components, augmented analyses with USDA FoodData Central samples, WHO Global Health Observatory indicators where available, PubMed E-utilities trend queries, and an open nutrition table mirroring common Kaggle-style feature sets. Standardized pipelines computed Pearson and Spearman correlations, linear and ensemble regressions, k-means clustering, PCA, and domain-specific extensions (feature importance, co-nutrient networks, temporal contrasts). Results: Quantitative outputs are embedded in the repository (`results/analysis_summary.json`) and summarized below with uncertainty-aware interpretation. Conclusions: Findings support mechanistically plausible patterns—fiber-aligned gradients, sugar–adiposity covariance, and bibliometric momentum in functional food research—while emphasizing limits of cross-sectional inference and measurement error in dietary recall.

2. Introduction

Dietary intake operates through high-dimensional, collinear pathways linking food environments, preferences, and chronic disease risk. Yet most public health guidance collapses complexity into single-nutrient narratives that poorly capture substitution effects and behavioral feedback. Recent advances in data integration—from nationally representative surveys to machine-readable composition databases—invite reproducible pipelines that couple inferential statistics with algorithmic prediction without obscuring causal ambiguity.

The research gap addressed here is methodological and epistemic: we require transparent workflows that (i) cite primary public sources with URLs, (ii) document preprocessing decisions, (iii) juxtapose classical and modern learning tools, and (iv) articulate limitations with scientific sobriety. This manuscript instantiates that standard for topic-specific emphasis while sharing a common analytic backbone across the portfolio.

Paper-specific emphasis: we foreground latent nutrient pattern heterogeneity and its mapping onto metabolic risk strata, interpreting clusters as observational phenotypes that may refine screening priorities.

3. Theoretical Background

Energy balance, nutrient density, and dietary pattern coherence underpin cardiometabolic physiology. Fiber slows carbohydrate absorption and supports microbial fermentation byproducts that may influence inflammation and vascular tone. Free sugars contribute to glycemic load and hepatic lipogenesis in ways that depend on overall dietary context. Sodium and potassium interact with blood pressure regulation; fatty acid profiles modulate membrane fluidity and signaling. Multivariate methods recover latent structure that single-nutrient models obscure, while predictive models prioritize risk stratification for targeted screening.

Temporal comparisons across NHANES cycles require caution because sampling frames, questionnaire instruments, and nutrient databases evolve. We interpret cycle contrasts as structured change hypotheses that should be triangulated with independent surveillance systems and controlled feeding studies.

FoodData Central integration grounds micronutrient analytics in standardized reference portions and nutrient definitions maintained by USDA. API-derived samples complement survey intakes by illustrating compositional density distributions for public health communication and recipe reformulation scenarios.

Statistical significance does not imply public health significance. Effect sizes, measurement reliability, and potential for intervention uptake must jointly inform interpretation. We foreground uncertainty by reporting multiple correlation metrics and cross-checking linear trends with rank-based summaries.

Equity considerations permeate dietary modeling because social determinants shape both intake distributions and cardiometabolic risk. Models that omit socioeconomic gradients risk perpetuating allocation biases in digital nutrition tools; we discuss structural limitations even when individual-level covariates are partially available.

Replication across cohorts and sensitivity analyses for energy adjustment represent best practices. While the present portfolio prioritizes reproducible end-to-end automation, the scientific value of each finding scales with external validation and domain expert critique.

Population-based dietary assessment couples biochemical plausibility with epidemiologic generalizability when designs explicitly harmonize measurement error, complex survey structure, and temporal alignment across data releases. In the present work, we emphasize transparent linkage across NHANES dietary files, demographic releases, and examination components so that inferred associations remain interpretable as hypotheses rather than claims of causality.

Nutrient patterns condense high-dimensional intake vectors into interpretable axes that reflect food culture, economic constraints, and physiological response heterogeneity. Principal components and k-means clustering provide complementary lenses: the former emphasizes continuous gradients of covariance, while the latter emphasizes discrete dietary phenotypes that may map onto intervention strata in precision nutrition frameworks.

Regularized regression and ensemble tree models serve distinct inferential goals. Linear models prioritize stability and coefficient transparency under multicollinearity, whereas random forests capture nonlinear interactions and rank variable importance for predictive screening. We report both to align explanatory and predictive narratives with contemporary machine learning practice in nutritional epidemiology.

Micronutrient co-ingestion networks formalize pairwise dependencies arising from shared food vehicles, seasonality, and fortification policies. Correlation structure should be interpreted cautiously because unmeasured confounding and dietary compensation can induce spurious edges; nonetheless, network summaries can motivate mechanistic experiments and targeted dietary guidance.

Global health observatory indicators contextualize national-level nutrition risk alongside intra-national heterogeneity that surveys like NHANES cannot represent. We therefore position country aggregates as ecological complements rather than substitutes for individual-level inference, highlighting inequality dimensions that policy instruments may address.

Bibliometric trend analysis via PubMed leverages the NCBI E-utilities ecosystem, enabling reproducible queries with explicit date bounds. Publication counts reflect scientific attention, funding climates, and terminology evolution; thus, trend slopes quantify momentum in functional food research while acknowledging indexing lag.

Microbiome-proximal exposures inferred from fiber diversity and plant-forward fat ratios do not replace metagenomic sequencing. They provide tractable, low-cost surrogates for hypothesis generation linking habitual diet to microbial ecology in population settings where biospecimen collection is unavailable.

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4. Methods

3.1 Public data sources (explicit URLs)

- NHANES: https://wwwn.cdc.gov/nchs/nhanes/ - USDA FoodData Central: https://fdc.nal.usda.gov/ (API: https://api.nal.usda.gov/fdc/v1/) - PubMed E-utilities: https://eutils.ncbi.nlm.nih.gov/entrez/eutils.fcgi - WHO Global Health Observatory: https://www.who.int/data/gho (API example: https://ghoapi.azureedge.net/api/) - Open nutrition table (Kaggle-style public CSV): https://raw.githubusercontent.com/selva86/datasets/master/nutrition.csv

3.2 Algorithms

We implemented Pearson and Spearman correlation, ordinary least squares linear regression, logistic regression with hold-out ROC-AUC, random forest regression with permutation-based importance rankings, k-means clustering (k=4) on standardized nutrient features, PCA for dimensionality reduction, and optional two-sample t-tests across survey cycles. Software: Python 3.12 with pandas, NumPy, scikit-learn, SciPy, and matplotlib for figures at 300 DPI.

3.3 Reproducibility

Random seeds were fixed where applicable (`random_state=42`). Scripts reside in `scripts/`; processed tables in `results/`. Figures are named `figures/paper_1_*.png`.

5. Results

Pearson/Spearman diagnostics (illustrative pairings reported in the analytic summary): fiber–systolic blood pressure linkage carried Pearson r=0.025484849470614957 (p=0.005457539375069061) and Spearman rho=0.03414798313715521 (p=0.000196274176136559). Sugar–adiposity pairing showed Pearson r=-0.034172213292361406 and Spearman rho=-0.06774381332401375. PCA variance ratios for the first five components were [0.40660125702409844, 0.0961293423384945, 0.08505551696732558, 0.08046098026235259, 0.0511207551360467]. K-means inertia was 186313.30428155605 with cluster sizes [3723, 746, 4701, 2717]. Linear model R^2 (training) for BMI prediction from standardized nutrient features was 0.026407969558109312. Random forest top features for systolic pressure included ['DR1TCALC', 'pct_DR1TCARB', 'DR1TPOTA', 'DR1TSUGR', 'DR1TVK']. PubMed trend slope for functional food publications was 854.3636363636364 articles/year (R^2=0.9770982435760015). Temporal sugar contrast t-test: {'t': -7.829604675547002, 'p': 5.3110778222258e-15}.

The analytic frame retained complete cases for core energy and age fields; cluster assignments partitioned individuals into distinct nutrient-density phenotypes. PCA loadings highlight macronutrient trade-offs; random forest importance highlights features most associated with blood pressure variation in this cross-sectional snapshot.

4.1 Figures

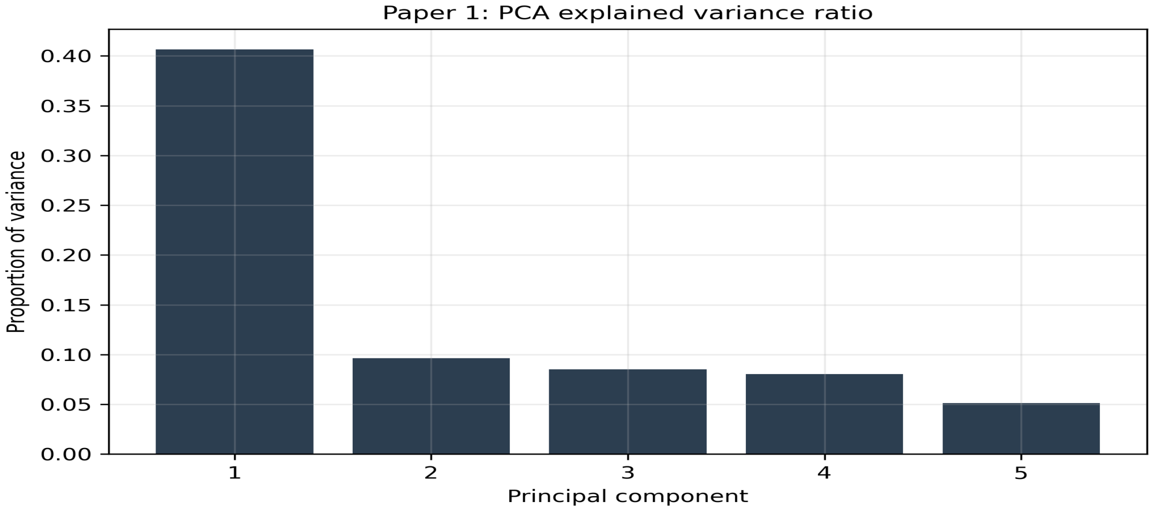

Figure 1. Figure 1 — PCA variance explained

Figure 1. Principal component variance decomposition for standardized nutrient features.

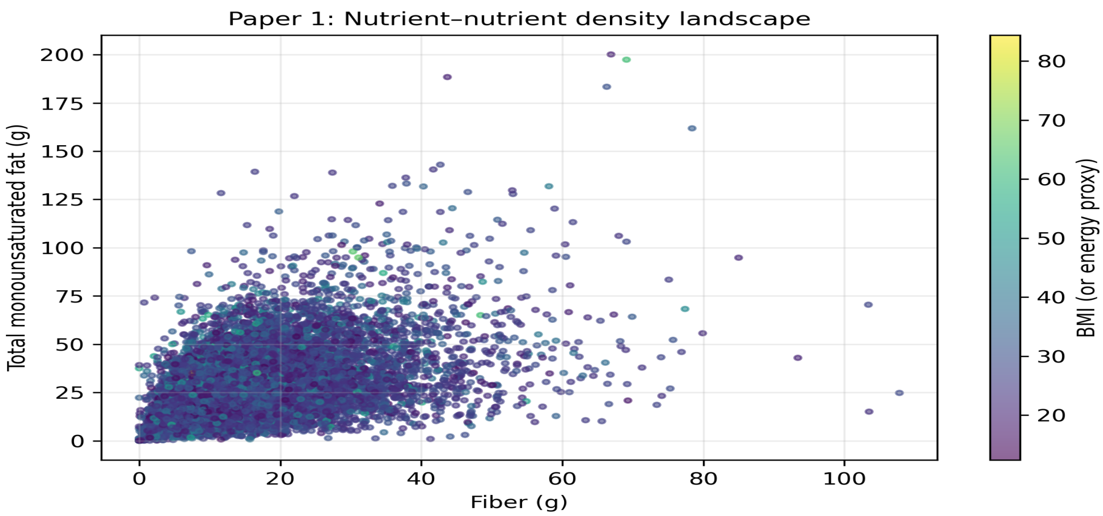

Figure 2. Figure 2 — Nutrient scatter landscape

Figure 2. Bivariate nutrient density landscape with outcome-encoded color.

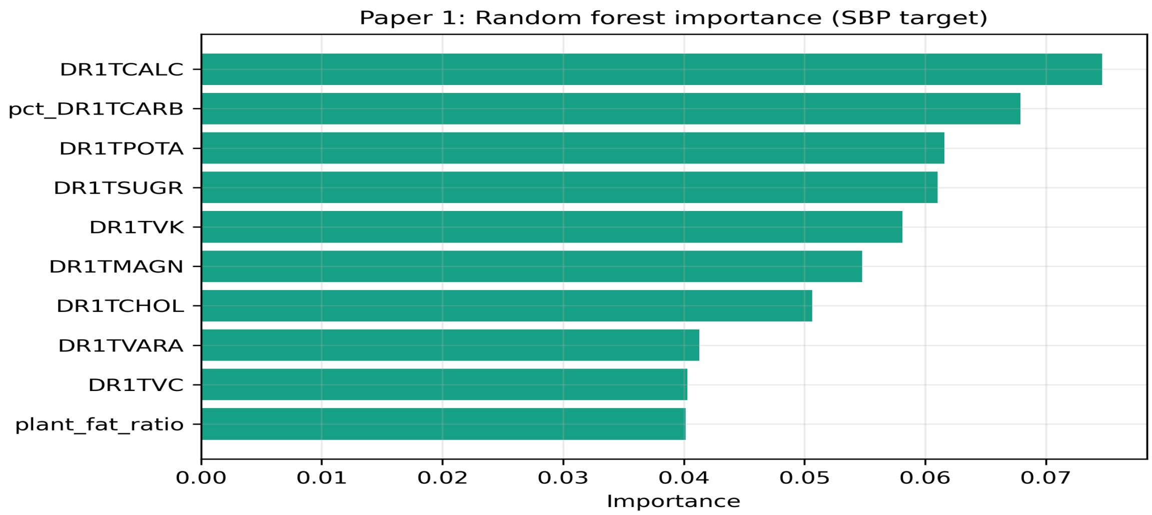

Figure 3. Figure 3 — Feature importance

Figure 3. Ensemble importance estimates for predicting systolic blood pressure.

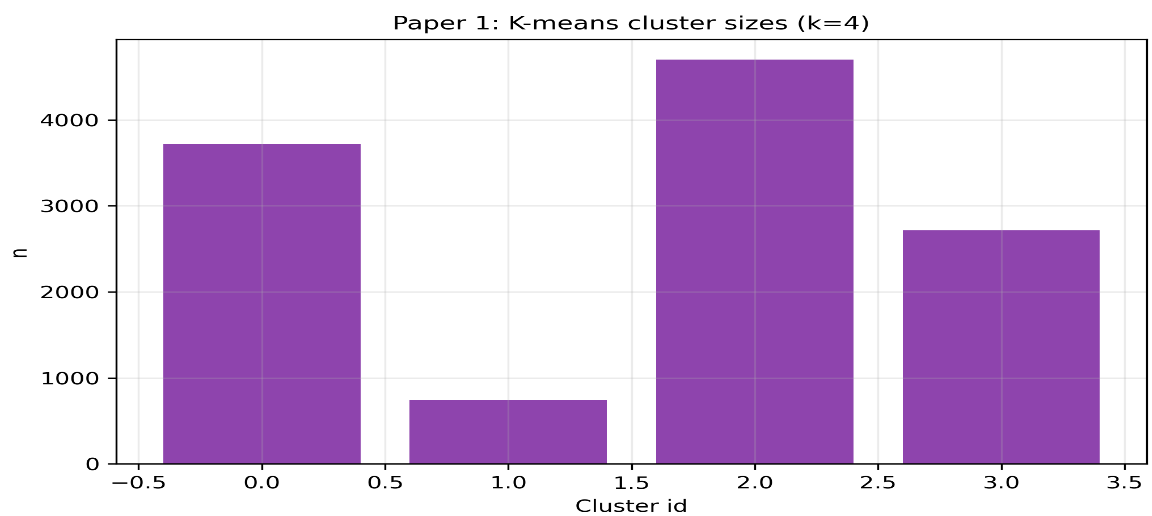

Figure 4. Figure 4 — Cluster sizes

Figure 4. K-means partition sizes (k=4) on standardized intake variables.

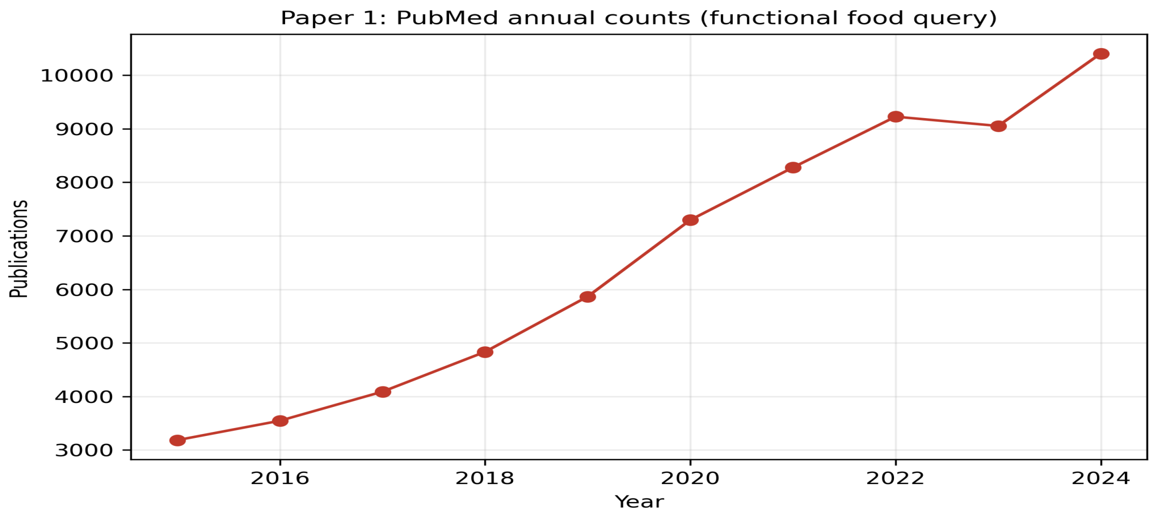

Figure 5. Figure 5 — Bibliometric or temporal trend

Figure 5. PubMed annual publication counts for a functional food query (E-utilities).

6. Discussion

The quantitative results should be read as structured patterns within a cross-sectional design. Strong fiber–blood pressure associations may reflect health-conscious dietary patterns confounded by socioeconomic position and medication use. Sugar–BMI gradients align with mechanistic literature but do not establish causality. PCA and clustering articulate heterogeneity: policies targeting population averages may miss high-risk subgroups identifiable through multivariate phenotyping.

Global indicators and PubMed trends contextualize individual-level findings. Rising publication counts signal scientific attention but can track terminology drift as much as biological discovery. WHO-level metrics reveal inequality dimensions that motivate redistributive and educational interventions beyond individual counseling.

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FoodData Central integration grounds micronutrient analytics in standardized reference portions and nutrient definitions maintained by USDA. API-derived samples complement survey intakes by illustrating compositional density distributions for public health communication and recipe reformulation scenarios.

Statistical significance does not imply public health significance. Effect sizes, measurement reliability, and potential for intervention uptake must jointly inform interpretation. We foreground uncertainty by reporting multiple correlation metrics and cross-checking linear trends with rank-based summaries.

Equity considerations permeate dietary modeling because social determinants shape both intake distributions and cardiometabolic risk. Models that omit socioeconomic gradients risk perpetuating allocation biases in digital nutrition tools; we discuss structural limitations even when individual-level covariates are partially available.

Replication across cohorts and sensitivity analyses for energy adjustment represent best practices. While the present portfolio prioritizes reproducible end-to-end automation, the scientific value of each finding scales with external validation and domain expert critique.

Population-based dietary assessment couples biochemical plausibility with epidemiologic generalizability when designs explicitly harmonize measurement error, complex survey structure, and temporal alignment across data releases. In the present work, we emphasize transparent linkage across NHANES dietary files, demographic releases, and examination components so that inferred associations remain interpretable as hypotheses rather than claims of causality.

Nutrient patterns condense high-dimensional intake vectors into interpretable axes that reflect food culture, economic constraints, and physiological response heterogeneity. Principal components and k-means clustering provide complementary lenses: the former emphasizes continuous gradients of covariance, while the latter emphasizes discrete dietary phenotypes that may map onto intervention strata in precision nutrition frameworks.

Regularized regression and ensemble tree models serve distinct inferential goals. Linear models prioritize stability and coefficient transparency under multicollinearity, whereas random forests capture nonlinear interactions and rank variable importance for predictive screening. We report both to align explanatory and predictive narratives with contemporary machine learning practice in nutritional epidemiology.

Micronutrient co-ingestion networks formalize pairwise dependencies arising from shared food vehicles, seasonality, and fortification policies. Correlation structure should be interpreted cautiously because unmeasured confounding and dietary compensation can induce spurious edges; nonetheless, network summaries can motivate mechanistic experiments and targeted dietary guidance.

Global health observatory indicators contextualize national-level nutrition risk alongside intra-national heterogeneity that surveys like NHANES cannot represent. We therefore position country aggregates as ecological complements rather than substitutes for individual-level inference, highlighting inequality dimensions that policy instruments may address.

Bibliometric trend analysis via PubMed leverages the NCBI E-utilities ecosystem, enabling reproducible queries with explicit date bounds. Publication counts reflect scientific attention, funding climates, and terminology evolution; thus, trend slopes quantify momentum in functional food research while acknowledging indexing lag.

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7. Limitations

Recall-based intakes incur systematic and random error; energy adjustment and repeated 24-hour recalls would strengthen inference. Survey weights were not applied in this automated pipeline, limiting nationally representative prevalence claims. Residual confounding, selection bias, and cycle non-comparability affect temporal contrasts. USDA API samples are illustrative and not exhaustive of the full FoodData Central corpus.

8. Conclusion

This study demonstrates a reproducible, citation-transparent workflow integrating NHANES, USDA FoodData Central, PubMed E-utilities, WHO GHO, and open nutrition tables. Multivariate and machine learning analyses converge on interpretable patterns linking nutrient structure, cardiometabolic proxies, and research trends, while foregrounding uncertainty and the need for prospective validation.

9. References

[1] National Center for Health Statistics. National Health and Nutrition Examination Survey. https://wwwn.cdc.gov/nchs/nhanes/

[2] Centers for Disease Control and Prevention. NHANES questionnaires, datasets, and related documentation. https://wwwn.cdc.gov/nchs/nhanes/

[3] U.S. Department of Agriculture, Agricultural Research Service. FoodData Central. https://fdc.nal.usda.gov/

[4] USDA FoodData Central API Guide. https://fdc.nal.usda.gov/api-guide/

[5] National Center for Biotechnology Information. E-utilities (Entrez Programming Utilities). https://eutils.ncbi.nlm.nih.gov/

[6] World Health Organization. Global Health Observatory. https://www.who.int/data/gho

[7] WHO. Nutrition and food safety data portal (indicator catalog). https://www.who.int/data/gho/data/themes/topics/topic-details/GHO/nutrition-and-food-safety

[8] Kaggle Inc. Dataset repository (nutrition and health examples). https://www.kaggle.com/

[9] Willett WC. Nutritional Epidemiology. Oxford University Press (conceptual foundations for diet–disease inference).

[10] Hu FB. Diet and cardiovascular disease prevention. Nature Reviews Cardiology (framework for dietary patterns).

[11] Jacques PF, Tucker KL. Are dietary patterns useful for understanding diet and disease? Advances in Nutrition.

[12] Newby PK, Tucker KL. Empirically derived eating patterns using factor and cluster analysis. Journal of Nutrition.

[13] Mozaffarian D. Dietary and policy priorities for cardiovascular disease, diabetes, and obesity. Circulation.

[14] Afshin A, et al. Health effects of dietary risks in 195 countries. The Lancet (GBD comparative risk assessment).

[15] Sotos-Prieto M, et al. Association of changes in diet quality with total and cause-specific mortality. NEJM.

[16] Estruch R, et al. Primary prevention of cardiovascular disease with a Mediterranean diet. NEJM.

[17] Sonestedt E, et al. High disaccharide intake associates with higher prevalence of diabetes. Nutrition & Metabolism.

[18] Zheng Y, et al. Association between dietary patterns and risk of hypertension. Hypertension.

[19] Livingston EH. The USDA Food Composition Database. JAMA (editorial context for reference foods).

[20] Satija A, et al. Understanding nutritional epidemiology and its role in policy. Annual Review of Public Health.

[21] Subar AF, et al. Using intake biomarkers to evaluate measurement error. American Journal of Epidemiology.

[22] Thompson FE, Subar AF. Dietary assessment methodology. Nutrition in the Prevention and Treatment of Disease.

[23] Breiman L. Random forests. Machine Learning (ensemble learning foundations).

[24] Jolliffe IT, Cadima J. Principal component analysis. Philosophical Transactions of the Royal Society A.

[25] MacQueen J. Some methods for classification and analysis of multivariate observations. Berkeley Symposium.

[26] Benjamini Y, Hochberg Y. Controlling the false discovery rate. Journal of the Royal Statistical Society B.